

WEEK 7 – TUTORIAL

Objectives:

The objective of this tutorial is to provide students with **practical understanding and hands-on experience in project resource management, including planning, estimating, acquiring, allocating, and developing resources for IT projects**. Students will learn to identify all necessary resources, create a Resource Breakdown Structure (RBS), analyse over- and under-allocated resources, implement resource leveling, and apply strategies to improve team productivity and morale using relevant motivation theories. By the end of the session, students should be able to effectively plan, manage, and optimize resources in real-world IT project scenarios.

Tutorial Activities:

- Resource Planning and Estimation
- Resource Acquisition and Estimation
- Team development and motivation

Learning outcomes

By the end of this tutorial, students will be able to:

- **LO1.** Understand each of the core project management knowledge areas
- **LO2.** Apply project management knowledge, tools and techniques
- **LO4.** Develop basic proficiency in project management tools and practices

Resource Planning and Estimation

For a Mobile Banking App, identify all resources needed and draw a simple RBS in a hierarchical format.

Potential list of resources:

- *Human Resources*
 - *Backend developers*
 - *Frontend developers*
 - *QA Testers*
 - *DevOps Engineer*
 - *Project Manager*
- *Equipment/Tools*
 - *Laptops/Workstations*
 - *Servers/Cloud infrastructure*
 - *Networking equipment*
- *Software/Materials*
 - *Development IDEs*
 - *Testing tools*
 - *Project management software*
 - *Documentation tools*

Resource Acquisition and Allocation

Identify over-allocated and underutilized resources. Furthermore, suggest a resource leveling plan to balance the workload.



Team Member	Task Hours Assigned	Availability
Dev 1	50	40
Dev 2	30	40
QA 1	35	40
QA 2	40	40

- Dev 1: Over-allocated by 10 hours (50 assigned vs 40 available)
- Dev 2: Underutilized by 10 hours (30 assigned vs 40 available)
- QA 1: Slightly underutilized (35 vs 40)
- QA 2: Fully allocated

Team Member	Task Hours Assigned	Availability	Status
Dev 1	40	40	Balanced
Dev 2	40	40	Balanced
QA 1	37.5	40	Slightly underutilized
QA 2	37.5	40	Slightly underutilized

Team development and Motivation

A QA team is underperforming due to low motivation. Suggest strategies to improve team productivity and morale. Furthermore, discuss which motivation theory applies and why

- Strategies to Improve Productivity and Morale:
 - Recognition and Rewards: Acknowledge QA achievements and provide incentives for meeting testing targets.
 - Training and Skill Development: Offer workshops to improve confidence and efficiency.
 - Workload Management: Ensure fair distribution of tasks to avoid burnout.
 - Goal Setting and Feedback: Set clear, achievable goals and provide regular feedback.
 - Team Building Activities: Foster collaboration and positive team culture.
 - Empowerment: Involve QA team in decision-making for testing strategies.
- Motivation theory applied:
 - Herzberg's Motivation-Hygiene Theory:
 - Hygiene factors (salary, workload, working conditions) are managed to prevent dissatisfaction.
 - Motivators (recognition, achievement, growth opportunities) are applied to increase performance and morale.
 - Alternatively, Maslow's Hierarchy of Needs can be used: ensure basic needs (salary, safe work environment) are met, then focus on esteem and self-actualization through recognition, training, and career growth.